DOMESTIC ELECTRICAL INSTALLATION CONDITION REPORT

Requirements For Electrical Installations - BS 7671 IET Wiring Regulations

Report Reference:	CE202695

1 DETAILS OF THE PERSON ORDERING THE REPORT

Client:	Highfield Hall Community Centre
Address:	Marsh Lane, Farnworth, Bolton

2 REASON FOR PRODUCING THIS REPORT

Reason for producing this report:	Safety assessment requested by client. To assess compliance with BS 7671 (2018) Amendment 2 (2022) and the Electricity at Work Regulations 1989.
Date(s) on which inspection and testing was carried out:	15/03/2024

3 DETAILS OF THE INSTALLATION WHICH IS THE SUBJECT OF THIS REPORT

Installation Address:	Marsh Lane, Farnworth, Bolton, BL4 0AW
Estimated age of wiring system:	15 years
Evidence of additions/alterations:	Yes
If yes, estimated age:	5 years
Installation records available? (Regulation 651.1)	No
Date of last inspection:	10/04/2019

4 EXTENT AND LIMITATIONS OF INSPECTION AND TESTING

Extent of the electrical installation covered by this report:	50% of the installation in accordance with item 3.8.4 of Guidance Note 3. Circuits in the main hall and offices were tested in full; storage and utility areas were visually inspected only.
Agreed limitations including the reasons (see Regulation 653.2):	No testing of HVAC control cables. No lifting of floor boards or inspection of buried conduit. Areas inaccessible due to ongoing building use were excluded from destructive testing.
Agreed with:	Mr. J. Hargreaves (Centre Manager)
Operational limitations including the reasons:	Building in use during inspection. Some circuits could not be de-energised for testing.

The inspection and testing detailed in this report and accompanying schedules have been carried out in accordance with BS 7671:2018 (IET Wiring Regulations) as amended to 2020. It should be noted that cables concealed within trunking and conduits, under floors, in roof spaces, and generally within the fabric of the building or underground, have not been inspected unless specifically agreed between the client and inspector prior to the inspection. An inspection should be made within an accessible roof space housing other electrical equipment.

5 SUMMARY OF THE CONDITION OF THE INSTALLATION

See page 3 for a summary of the general condition of the installation in terms of electrical safety.

Overall assessment of the installation in terms of its suitability for continued use*:	**SATISFACTORY**

** An unsatisfactory assessment indicates that dangerous (Code C1) and/or potentially dangerous (Code C2) conditions have been identified.*

6 RECOMMENDATIONS

Where the overall assessment of the suitability of the installation for continued use on page 1 is stated as 'UNSATISFACTORY', I/We recommend that any observations classified as 'Code 1 - Danger Present' or 'Code 2 - Potentially dangerous' are acted upon as a matter of urgency. Investigation without delay is recommended for observations identified as 'FI - Further Investigation Required'. Observations classified as 'Code 3 - Improvement recommended' should be given due consideration.

Subject to the necessary remedial action being taken, I/we recommend that the installation is further inspected and tested by:	3 Years

Note: The proposed date for the next inspection should take into consideration the frequency and quality of maintenance that the installation can reasonably be expected to receive during its intended life. The period should be agreed between relevant parties.

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7 OBSERVATIONS AND RECOMMENDATIONS FOR ACTIONS TO BE TAKEN

Referring to the attached schedules of inspection and test results, and subject to the limitations specified on page 1 of this report under 'Extent of the Installation and Limitations of Inspection and Testing':

Item No	Observations	Classification Code
1	Inspection Schedule Item 1.1: The means of isolation (main switch) does not have provision for securing in the off position. Regulation 537.3.2 refers.	C2
2	Inspection Schedule Item 3.4: Conductor insulation is discoloured and shows signs of thermal damage at the connection to luminaire fitting in office 3. Regulation 512.1.1 refers.	C2
3	Inspection Schedule Item 6.2: IP rating of luminaires is not appropriate for Zone 1 in the kitchen area. IP2X or IPXXB minimum is required. Regulation 701.512.2 refers.	C3
4	Inspection Schedule Item 8.1: The external consumer unit cover does not close securely, allowing access to live parts. Regulation 416.1 refers.	C2
5	Inspection Schedule Item 4.3: Single-phase socket outlets installed without addition [illegible] io location where use of portable equipment outdoors is reasonably foreseeable. Regulation 411.3.3 refers.	C3
6	Inspection Schedule Item 5.9: Cables in the ceiling void are not adequately supported and are resting on suspended ceiling tiles. Risk of mechanical damage and overheating. Regulation 522.8.5 refers.	C3
7	Inspection Schedule Item 2.7: Earth fault loop impedance for circuit 14 (HVAC) measured at 1.92Ω exceeds the maximum value of 1.84Ω permitted for the installed 16A Type B MCB. Further investigation required to determine cause.	FI

or

The following observations and recommendations are made

One of the following codes, as appropriate, has been allocated to each of the observations made above to indicate to the person(s) responsible for the installation the degree of urgency for remedial action.

C1	C2	C3	FI
Danger Present Risk of injury. Immediate remedial action required	**Potentially dangerous** Urgent remedial action required	**Improvement recommended**	Further investigation required without delay

Immediate remedial action required for items:	N/A
Urgent remedial action required for items:	1, 2, 3
Improvement recommended for items:	1, 2, 3
Further investigation required for items:	1

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8 GENERAL CONDITION OF THE INSTALLATION

General condition of the installation (in terms of electrical safety):	The installation was found to be in a generally good condition, adequate for continued use. A number of C3 observations were made that should be attended to at the earliest opportunity to ensure continued safety.

9 DECLARATION FOR THE INSPECTION, TESTING AND ASSESSMENT

I/We, being the person(s) responsible for the inspection and testing of the electrical installation (as indicated by my/our signatures below), particulars of which are described above, having exercised reasonable skill and care when carrying out the inspection and testing, hereby declare that the information in this report, including the observations and the attached schedules, provides an accurate assessment of the condition of the electrical installation taking into account the stated extent and limitations in section 4 of this report.

Trading Title:	Cain Enabled Engineering Ltd	**Address:**	Piccadilly Business Centre, 8 Piccadilly, Manchester, M12 6AE
Registration Number (if applicable):	611716000	**Telephone Number:**	01246 387 450

For the INSPECTION, TESTING AND ASSESSMENT of the report:

Name:	Daniel Allport	**Position:**	Qualified Supervisor
Signature:		**Date:**	15/03/2024

10 DETAILS OF TEST INSTRUMENTS USED

Details of test instruments used (state serial and/or asset numbers):

Multi-functional:		**Earth electrode resistance:**	N/A
Insulation resistance:	N/A	**Earth fault loop impedance:**	N/A
Continuity:	N/A	**RCD:**	N/A

11 SUPPLY CHARACTERISTICS AND EARTHING ARRANGEMENTS AT THE ORIGIN

Earthing Arrangement

[ ] TN-S
[X] TN-C-S
[ ] TT
Other:

Number and Type of Live Conductors

[ ] 1-phase (2 wire):
[ ] 3-phase (3 wire):
[ ] 1-phase (3 wire):
[X] 3-phase (4 wire):

Confirmation of supply polarity: Yes

Supply Parameters

Nominal voltage(s): U:	400 V
Nominal voltage(s): Uo:	230 V
Nominal frequency, f:	50 Hz
Prospective fault current, Ipf:	1.8 kA
External earth fault loop impedance, Ze:	0.13 Ω

Distributor's Protective Device

BS(EN):	BS 1361
Type:	BS 1361 Fuse HBC
Rated current:	100 A
Short-circuit capacity:	33 kA

12 PARTICULARS OF INSTALLATION REFERRED TO IN THIS REPORT

Means of Earthing [X] Distributor's facility [ ] Installation earth electrode

Earth Electrode Details (where applicable)

Type:	N/A
Resistance to Earth:	N/A
Location:	N/A

Maximum Demand / Protective Measures

Maximum Demand (Load):	100 Amps
Protective measure(s):	ADS (Automatic Disconnection of Supply)
Method of measurement:	N/A

Main Switch / Switch-Fuse / Circuit-Breaker / RCD

Type BS(EN):	60947-3 Isolator
Number of poles:	2
Current rating:	100 A
Fuse/device rating:	100 A
Voltage rating:	240 V
Supply conductors:	Copper
Supply conductors csa:	25 mm²

RCD Main Switch Details

Rated residual current (I∆n):	N/A
Rated time delay:	N/A
Measured operating time:	N/A

Earthing Conductor

Material:	Copper
CSA:	16 mm²
Verified:	Yes

Main Protective Bonding Conductor

Material:	Copper
CSA:	10 mm²
Verified:	Yes

Bonding to Extraneous-Conductive-Parts

Water installation pipes:	Yes
Gas installation pipes:	Yes
Oil installation pipes:	N/A
Lightning protection:	N/A
Structural steel:	N/A

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13 INSPECTION SCHEDULE FOR DOMESTIC & SIMILAR PREMISES WITH UP TO 100A SUPPLY

Item	Description	Comments	Outcome
1.0	**EXTERNAL CONDITION OF INTAKE EQUIPMENT (VISUAL INSPECTION ONLY)**		
1.1	Service cable		'
1.2	Service head		'
1.3	Earthing arrangement		'
1.4	Meter tails		'
1.5	Metering equipment		'
1.6	Isolator (where present)		'
2.0	**PRESENCE OF ADEQUATE ARRANGEMENTS FOR OTHER SOURCES SUCH AS MICROGENERATORS (551.6; 551.7)**		N/A
3.0	**EARTHING / BONDING ARRANGEMENTS (411.3; Chap 54)**		
3.1	Presence and condition of distributor's earthing arrangement (542.1.2.1; 542.1.2.2)		'
3.2	Presence and condition of earth electrode connection where applicable (542.1.2.3)		'
3.3	Provision of earthing/bonding labels at all appropriate locations (514.13.1)		'
3.4	Confirmation of earthing conductor size (542.3; 543.1.1)		'
3.5	Accessibility and condition of earthing conductor at MET (543.3.2)		'
3.6	Confirmation of main protective bonding conductor sizes (544.1)		'
3.7	Condition and accessibility of main protective bonding conductor connections (543.3.2; 544.1.2)		'
3.8	Accessibility and condition of other protective bonding connections (543.3.1; 543.3.2)		'
4.0	**CONSUMER UNIT(S) / DISTRIBUTION BOARD(S)**		
4.1	Adequacy of working space/accessibility to consumer unit/distribution board (132.12; 513.1)		'
4.2	Security of fixing (134.1.1)		'
4.3	Condition of enclosure(s) in terms of IP rating etc (416.2)		'
4.4	Condition of enclosure(s) in terms of fire rating etc (421.1.201; 526.5)		'
4.5	Enclosure not damaged/deteriorated so as to impair safety (651.2)		'
4.6	Presence of main linked switch (as required by 462.1.201)		'
4.7	Operation of main switch (functional check) (643.10)		'
4.8	Manual operation of circuit-breakers and RCDs to prove disconnection (643.10)		'
4.9	Correct identification of circuit details and protective devices (514.8.1; 514.9.1)		'
4.10	Presence of RCD six-monthly test notice at or near consumer unit/distribution board (514.12.2)		'
4.11	Presence of non-standard (mixed) cable colour warning notice at or near consumer unit/distribution board (514.14)		'
4.12	Presence of alternative supply warning notice at or near consumer unit/distribution board (514.15)		'
4.13	Presence of other required labelling (please specify) (Section 514)		'
4.14	Compatibility of protective devices, bases and other components; correct type and rating (No signs of unacceptable thermal damage, arcing or overheating) (411.3.2; 411.4; 411.5; 411.6; Sections 432, 433)		'
4.15	Single-pole switching or protective devices in line conductor only (132.14.1; 530.3.3)		'
4.16	Protection against mechanical damage where cables enter consumer unit/distribution board (132.14.1; 522.8.1; 522.8.5; 522.8.11)		'
4.17	Protection against electromagnetic effects where cables enter consumer unit/distribution board/enclosures (521.5.1)		'
4.18	RCD(s) provided for fault protection - includes RCBOs (411.4.204; 411.5.2; 531.2)		'
4.19	RCD(s) provided for additional protection/requirements - includes RCBOs (411.3.3; 415.1)		'
4.20	Confirmation of indication that SPD is functional (651.4)		'
4.21	Confirmation that ALL conductor connections, including connections to busbars, are correctly located in terminals and are tight and secure (526.1)		'
4.22	Adequate arrangements where a generating set operates as a switched alternative to the public supply (551.6)		'
4.23	Adequate arrangements where a generating set operates in parallel with the public supply (551.7)		'

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14 INSPECTION SCHEDULE FOR DOMESTIC & SIMILAR PREMISES WITH UP TO 100A SUPPLY

Item	Description	Comments	Outcome
5.0	**FINAL CIRCUITS**		
5.1	Identification of conductors (514.3.1)		'
5.2	Cables correctly supported throughout their run (521.10.202; 522.8.5)		'
5.3	Condition of insulation of live parts (416.1)		'
5.4	Non-sheathed cables protected by enclosure in conduit, ducting or trunking (521.10.1)		'
5.4.1	To include the integrity of conduit and trunking systems (metallic and plastic)		'
5.5	Adequacy of cables for current-carrying capacity with regard for the type and nature of installation (Section 523)		'
5.6	Coordination between conductors and overload protective devices (433.1; 533.2.1)		'
5.7	Adequacy of protective devices: type and rated current for fault protection (411.3)		'
5.8	Presence and adequacy of circuit protective conductors (411.3.1; Section 543)		'
5.9	Wiring system(s) appropriate for the type and nature of the installation and external influences (Section 522)		'
5.10	Concealed cables installed in prescribed zones (see Section 4. Extent and Limitations) (522.6.202)		'
5.11	Cables concealed under floors, above ceilings or in walls/partitions, adequately protected against damage (see Section 4. Extent and Limitations) (522.6.204)		'
5.12	Provision of additional requirements for protection by RCD not exceeding 30mA:		'
5.12.1	For all socket-outlets of rating 32A or less, unless an exception is permitted (411.3.3)		'
5.12.2	For the supply of mobile equipment not exceeding 32A rating for use outdoors (411.3.3)		'
5.12.3	For cables concealed in walls at a depth of less than 50mm (522.6.202; 522.6.203)		'
5.12.4	For cables concealed in walls/partitions containing metal parts regardless of depth (522.6.203)		'
5.12.5	Final circuits supplying luminaires within domestic (household) premises (411.3.4)		'
5.13	Provision of fire barriers, sealing arrangements and protection against thermal effects (Section 527)		'
5.14	Band II cables segregated/separated from Band I cables (528.1)		'
5.15	Cables segregated/separated from communications cabling (528.2)		'
5.16	Cables segregated/separated from non-electrical services (528.3)		'
5.17	Termination of cables at enclosures - indicate extent of sampling in Section 4 of the report (Section 526)		'
5.17.1	Connections soundly made and under no undue strain (526.6)		'
5.17.2	No basic insulation of a conductor visible outside enclosure (526.8)		'
5.17.3	Connections of live conductors adequately enclosed (526.5)		'
5.17.4	Adequately connected at point of entry to enclosure (glands, bushes etc.) (522.8.5)		'
5.18	Condition of accessories including socket-outlets, switches and joint boxes (651.2(v))		'
5.19	Suitability of accessories for external influences (512.2)		'
5.20	Adequacy of working space/accessibility to equipment (132.12; 513.1)		'
5.21	Single-pole switching or protective devices in line conductors only (132.14.1, 530.3.3)		'

15 INSPECTION SCHEDULE FOR DOMESTIC & SIMILAR PREMISES WITH UP TO 100A SUPPLY

Item	Description	Comments	Outcome
6.0	**LOCATION(S) CONTAINING A BATH OR SHOWER**		
6.1	Additional protection for all low voltage (LV) circuits by RCD not exceeding 30mA (701.411.3.3)		'
6.2	Where used as a protective measure, requirements for SELV or PELV met (701.414.4.5)		'
6.3	Shaver sockets comply with BS EN 61558-2-5 formerly BS 3535 (701.512.3)		'
6.4	Presence of supplementary bonding conductors, unless not required by BS 7671:2018 (701.415.2)		'
6.5	Low voltage (e.g. 230 volt) socket-outlets sited at least 3m from zone 1 (701.512.3)		'
6.6	Suitability of equipment for external influences for installed location in terms of IP rating (701.512.2)		'
6.7	Suitability of accessories and controlgear etc. for a particular zone (701.512.3)		'
6.8	Suitability of current-using equipment for particular position within the location (701.55)		'
7.0	**OTHER PART 7 SPECIAL INSTALLATIONS OR LOCATIONS**		
7.1	N/A		**N/A**
7.2	N/A		**N/A**
7.3	N/A		**N/A**
7.4	N/A		**N/A**
7.5	N/A		**N/A**
7.6	N/A		**N/A**
7.7	N/A		**N/A**
7.8	N/A		**N/A**
7.9	N/A		**N/A**
7.10	N/A		**N/A**

OUTCOME CODES ' Acceptable C1 Danger present C2 Potentially dangerous C3 Improvement recommended FI Further investigation NV N

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16 SCHEDULE OF CIRCUIT DETAILS AND TEST RESULTS

Designation: D.B.1	**Location:** Meter Cupboard	**Prospective fault current (Ipf):** 1.8 kA

Circuit number	Circuit designation	Type of wiring	Ref. method	No. of points	Circuit conductors: csa		Max disc. time (s)	Overcurrent protective devices				RCD	Max Zs perm. (Ω)	Ring final circuits only			Circuit impedances: all circuits		Insulation resistance			Pol.	Max Zs measured (Ω)	RCD		AFDD
					Live (mm²)	cpc (mm²)		BS(EN)	Type	Rating (A)	Cap. (kA)	RCD IΔn (mA)		r1 (Line)	rn (Neutral)	r2 (cpc)	R1+R2 (Ω)	R2 (Ω)	Live-Live (MΩ)	Live-Earth (MΩ)	Test voltage (V)			Disc. time (ms)	RCD test btn	AFDD test btn
DB1-01	Main Lighting Rad…	A	C		1.5	1.5	0.4	60898	MCB Type B	32			1.44Ω				0.28	0.15					1.15			
DB1-02	Kitchen Ring Sock…	A	C		2.5	2.5	0.4	61009	RCBO Type C	16		30mA	1.92Ω				0.32	0.18					1.75			
DB1-03	Utility Radial So…	A	C		2.5	1.0	0.4	60269	BS88 Fuse	20			2.30Ω				0.45	0.22					2.65			
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CODES FOR TYPE OF WIRING

A Thermoplastic insulated/sheathed cables
B Thermoplastic cables in metallic conduit
C Thermoplastic cables in nonmetallic conduit
D Thermoplastic cables in metallic trunking
E Thermoplastic cables in nonmetallic trunking
F Thermoplastic/SWA cables
G Thermosetting/SWA cables
H Mineral insulated cables
O Other

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DOMESTIC ELECTRICAL INSTALLATION CONDITION REPORT

GUIDANCE FOR RECIPIENTS

(to be appended to the Report)

This Report is an important and valuable document which should be retained for future reference.

1. The purpose of this Report is to confirm, so far as reasonably practicable, whether or not the electrical installation is in a satisfactory condition for continued service (see Section 5). The Report should identify any damage, deterioration, defects and/or conditions which may give rise to danger.

2. The person ordering the Report should have received the 'original' Report and the inspector should have retained a duplicate.

3. The 'original' Report should be retained in a safe place and be made available to any person inspecting or undertaking work on the electrical installation in the future. If the property is vacated, this Report will provide the new owner/occupier with details of the condition of the electrical installation at the time the Report was issued.

4. Where the installation incorporates a residual current device (RCD) there should be a notice at or near the device stating that it should be tested six-monthly. For safety reasons it is important that this instruction is followed.

5. Section 4 (Extent and Limitations) should identify fully the extent of the installation covered by this Report and any limitations on the inspection and testing. The inspector should have agreed these aspects with the person ordering the Report and with other interested parties (licensing authority, insurance company, mortgage provider and the like) before the inspection was carried out.

6. Some operational limitations such as inability to gain access to parts of the installation or an item of equipment may have been encountered during the inspection. The inspector should have noted these in Section 4.

7. For items classified in Section 7 as C1 ('Danger present'), the safety of those using the installation is at risk, and it is recommended that a skilled person or persons competent in electrical installation work undertakes the necessary remedial work immediately.

8. For items classified in Section 7 as C2 ('Potentially dangerous'), the safety of those using the installation may be at risk and it is recommended that a skilled person or persons competent in electrical installation work undertakes the necessary remedial work as a matter of urgency.

9. Where it has been stated in Section 7 that an observation requires further investigation (code FI) the inspection has revealed an apparent deficiency which may result in a code C1 or C2, and could not, due to the extent or limitations of the inspection, be fully identified. Such observations should be investigated without delay. A further examination of the installation will be necessary, to determine the nature and extent of the apparent deficiency (see Section 6).

10. For safety reasons, the electrical installation should be re-inspected at appropriate intervals by a skilled person or persons, competent in such work. The recommended date by which the next inspection is due is stated in Section 6 of the Report under 'Recommendations' and on a label at or near to the consumer unit/distribution board.

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